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We formulate the Quantum Optimal Transport Amplitude Method (QOTAM) as an action-based information-projection representation of regulated quantum field theory. For positive Euclidean bosonic measures, the established Gibbs variational identity relates an unconstrained entropy minimum to the generating functional. Prescribed boundary marginals define a distinct Schrödinger bridge whose static cost follows from the field-theory endpoint law. We derive the relation between these two ensembles and identify the reweighting required to recover the original correlators. Detailed calculations state the admissibility conditions, source conventions, and connection to analytic continuation and scattering reduction. For fermions, a finite-matrix relative-entropy projection is formulated after reflection-positive Hamiltonian reconstruction, without assuming a positive classical fermion determinant or identifying a Gibbs state with a scattering process. A three-site interacting scalar calculation directly minimizes the full entropy functional and agrees with independent adaptive integration, including at couplings where the tested first- and second-order moment expansions are inaccurate. Its constrained bridge provides a separate test of marginal conservation and reweighting. Scalar and Yukawa tree responses and multidimensional packet tests are retained with their numerical scope stated explicitly. The construction is exact at finite regulator under the specified assumptions; computational advantage, continuum scattering, and a general fermionic process interpretation are not established by these tests.

I. INTRODUCTION

Scattering calculations require a dynamical theory, a prescription for extracting correlation functions, and a numerical representation of the resulting quantities. Optimal transport addresses a different question: how to couple specified probability measures subject to a cost or a reference law [1, 2]. Relating these questions requires distinguishing a positive coupling from a complex scattering amplitude. In particular, conservation of transport marginals is not operator unitarity, and an endpoint probability distribution does not specify the phase of a quantum wavepacket.

The Quantum Optimal Transport Amplitude Method (QOTAM) is developed here as an action-based information-projection representation of a regulated theory. The starting point is the Euclidean path integral, not a fitted amplitude. A normalized positive path law admits the standard Gibbs relative-entropy identity. Prescribed boundary distributions give a Schrödinger problem with that path law as its reference. The connection between dynamic and static Schrödinger problems and entropic transport is established mathematics [3, 4]; it is used here, rather than claimed as a new transport theorem. The original field-theory correlators are obtained from the *unconstrained* source-dependent variational value. Arbitrary constrained bridges generally describe different ensembles.

This formulation has two purposes. First, it specifies the input data, admissible measures, and relation of the variational values to kernels and correlators in one set of conventions. Second, it separates the numerical tasks that are sometimes combined under the name QOTAM: interacting free-energy minimization, boundary entropy projection, and reconstruction of action-derived scattering responses. A finite three-site interacting scalar example directly solves the first two problems without a coupling expansion, compares observables with independent adaptive quadrature, and tests them against first- and second-order perturbative truncations on the same field domain. Scalar and Yukawa tree calculations then check the scattering conventions, while a three-channel spectral calculation tests response functions on multidimensional packet samples. These are different tests, not interchangeable proofs of a single numerical algorithm.

For fermions, positivity of a Grassmann determinant is not assumed. We describe the finite-matrix entropy projection available after a suitable Euclidean reflection-positive theory has been reconstructed. The overlap chiral-Yukawa construction of Kikukawa and Usui provides a specified interacting example of the reflection-positivity input [5]. Reconstruction hypotheses and spectral cutoffs remain explicit. Quantum relative entropy and noncommutative entropic transport supply the associated operator setting [6, 7].

The contribution is the organized field-theory representation and its executable checks, including a direct interacting entropy minimization. It is not a new proof of the Gibbs principle, a derivation of QFT from transport axioms, or a demonstrated computational speedup. Core calculations are given in Appendix A; additional steps are collected in the accompanying *Mathematical derivations appendix*; every main-text result below states its assumptions and identifies the relevant established input.

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II. REGULATED ACTION AND THE GIBBS REPRESENTATION

A. Measure, conventions, and admissibility

A finite bosonic regulator leaves coordinates $\Phi \in \mathbb{R}^{d_N}$. Choose a source-independent positive quadratic reference and write

$$S_{E,N}(\Phi) = \tfrac{1}{2}\Phi^{\mathsf{T}} A_N \Phi + V_N(\Phi), \qquad A_N = A_N^{\mathsf{T}} > 0. \quad (1)$$

This is a reference decomposition, not an assertion that every interacting Hessian is positive. Gaussian integration defines a probability measure $\mathrm{d}R_{0,N} = Z_{0,N}^{-1} e^{-\Phi^{\mathsf{T}} A_N \Phi/(2\hbar)} \mathrm{d}\Phi$, where $Z_{0,N} = (2\pi\hbar)^{d_N/2}/\sqrt{\det A_N}$ and the covariance is $\hbar A_N^{-1}$. The normalization and source completion of the square are derived in Supplementary Appendix S.1.

For a real source J, set $U_J = V_N - J\cdot\Phi$ and define the exponential tilt

$$\mathcal{Z}_N[J] = \mathbb{E}_{R_{0,N}} e^{-U_J/\hbar}, \qquad \frac{\mathrm{d}P_{N,J}}{\mathrm{d}R_{0,N}} = \mathcal{Z}_N[J]^{-1} e^{-U_J/\hbar}. \quad (2)$$

Assume U_J is finite almost everywhere, $0 < \mathcal{Z}_N[J] < \infty$, and $\mathbb{E}_{P_{N,J}}|U_J| < \infty$. Trial probabilities satisfy $Q \ll R_{0,N}$, $D_{\mathrm{KL}}(Q\|R_{0,N}) < \infty$, and $\mathbb{E}_Q|U_J| < \infty$. Finite dimension alone does not imply these integrability conditions. A positive quartic leading term supplies them for the finite polynomial scalar models used here. Logarithms are natural and $\hbar > 0$ is retained until the scattering examples.

B. Variational value and its meaning

On this domain the Gibbs representation is

$$\mathfrak{F}_N[Q;J] = \mathbb{E}_Q U_J + \hbar D_{\mathrm{KL}}(Q\|R_{0,N}). \quad (3)$$

This is the familiar free-energy relative-entropy principle, with the objective measured in units of action. It is consistent with the entropy conventions used for Schrödinger problems [3]; stochastic control versions require additional hypotheses [8]. Taking the logarithm of Eq. (2) gives

$$\mathfrak{F}_N[Q;J] = -\hbar \log \mathcal{Z}_N[J] + \hbar D_{\mathrm{KL}}(Q\|P_{N,J}). \quad (4)$$

Nonnegativity of relative entropy proves the following specialization of the Gibbs principle; Supplementary Appendix S.2 supplies the variation and admissibility details.

Proposition 1 (Unconstrained variational value)**.** *Under the assumptions of Eq.* (2)*,*

$$\mathfrak{V}_N[J] := \inf_Q \mathfrak{F}_N[Q;J] = -\hbar \log \mathcal{Z}_N[J], \qquad Q_\star = P_{N,J}. \quad (5)$$

The minimizer is unique. If two source-independent reference–interaction splits represent the same microscopic action, their objectives differ on their common domain by a constant independent of Q and J.

The last assertion follows by retaining the free normalization factors: $Z_{0,A}\mathcal{Z}_A[J] = Z_{0,B}\mathcal{Z}_B[J]$. This is reference-split invariance. It is not uniqueness among all conceivable objective functions with the same minimizer. Multiplying the full objective by a positive constant also leaves the minimizer unchanged. Accordingly, $\varepsilon = \hbar$ below records the chosen action-unit normalization of the entropy representation, not a new independent physical prediction.

Equation (4) also identifies what a free-energy error measures: its excess above the exact minimum is $\hbar D_{\mathrm{KL}}(Q\|P_{N,J})$. This statement concerns the exact objective and exact minimum; an approximate entropy estimate or an uncertain reference value must carry its own numerical error. No error bound for on-shell scattering follows without further control of correlators, continuation, and reduction.

C. Source derivatives and kernel values

The connected generator is $W_{E,N}[J] = \hbar \log \mathcal{Z}_N[J] = -\mathfrak{V}_N[J]$, up to a source-independent normalization. For sources with a common integrable domination, differentiation gives

$$\begin{aligned} -\partial_{J_i}\mathfrak{V}_N &= \mathbb{E}_{P_{N,J}}\Phi_i, \\ -\hbar\partial_{J_i}\partial_{J_j}\mathfrak{V}_N &= \mathrm{Cov}_{P_{N,J}}(\Phi_i,\Phi_j), \\ G^{(n)}_{c,E;i_1\ldots i_n} &= -\hbar^{n-1}\partial_{J_{i_1}}\cdots\partial_{J_{i_n}}\mathfrak{V}_N. \end{aligned} \quad (6)$$

The logarithm selects connected cumulants; Gaussian contractions give the usual perturbative organization [9]. Supplementary Appendix S.4 derives these identities and distinguishes a formal coupling expansion from a convergent series. The exact identity itself makes no weak-coupling expansion.

For a positive Euclidean propagation kernel, normalized conditional free bridges R_0^{xy} give the fixed-endpoint version

$$\begin{aligned} \mathfrak{V}_S(x,y;T) &= \inf_{Q^{xy}}\{\mathbb{E}_{Q^{xy}}V_N + \hbar D_{\mathrm{KL}}(Q^{xy}\|R_0^{xy})\}, \\ K_E^S(y,x;T) &= K_{0,E}(y,x;T)e^{-\mathfrak{V}_S(x,y;T)/\hbar}. \end{aligned} \quad (7)$$

It follows by applying Eq. (4) to the ratio of the two fixed-boundary integrals. Continuous endpoints are understood through regular conditional kernels, not point masses assumed to have finite entropy relative to an unconditioned law. A normalized endpoint law additionally contains its boundary weights and normalization; it is not automatically the propagation kernel.

III. BOUNDARY PROJECTION AND ENTROPIC TRANSPORT

Let $B_\pm$ be specified measurable boundary maps on standard Borel spaces, put $P = P_{N,0}$, and let $R_S = (B_-, B_+)_\# P$ be their joint law. The prescribed probabilities $\rho_\pm$ define

$$\mathcal{A}(\rho_-, \rho_+) = \{Q : (B_-)_\# Q = \rho_-,\ (B_+)_\# Q = \rho_+\}. \quad (8)$$

The symbol # denotes push-forward. Restricting the same objective gives the Schrödinger information projection

$$Q_\star = \operatorname*{arg\,min}_{Q \in \mathcal{A}(\rho_-, \rho_+)} D_{\mathrm{KL}}(Q\|P). \quad (9)$$

This is an established path-space problem [3, 10, 11]. Its boundary data represent selected Euclidean configurations or mode statistics. Their identification with physical preparation/detection probabilities requires a measurement prescription; pure-state phases are separate input.

For $\gamma = (B_-, B_+)_\# Q$ and conditional laws Q^{xy}, P^{xy}, relative-entropy disintegration yields

$$D_{\mathrm{KL}}(Q\|P) = D_{\mathrm{KL}}(\gamma\|R_S) + \int D_{\mathrm{KL}}(Q^{xy}\|P^{xy})\,\mathrm{d}\gamma(x,y). \quad (10)$$

This standard chain rule and its application to Schrödinger bridges are given in Ref. [3]; Supplementary Appendix S.3 derives the Radon–Nikodym factorization. The second term is minimized by retaining the original conditional paths. Thus

$$\inf_{Q \in \mathcal{A}} D_{\mathrm{KL}}(Q\|P) = \inf_{\gamma \in \Pi(\rho_-, \rho_+)} D_{\mathrm{KL}}(\gamma\|R_S). \quad (11)$$

Here $\Pi(\rho_-, \rho_+)$ is the ordinary set of two-marginal couplings [1]. Uniqueness follows from strict convexity whenever a finite minimizer exists. In a finite discrete system with strictly positive reference entries and marginals, compactness gives existence, the optimizer is strictly positive, and its scaling form is

$$\gamma_{ij} = a_i R_{ij} b_j, \qquad a_i = \frac{r_i}{\sum_j R_{ij} b_j}, \quad b_j = \frac{s_j}{\sum_i a_i R_{ij}}. \quad (12)$$

These are the standard Sinkhorn/Schrödinger equations [4, 12], with no freedom to fit a scattering amplitude.

Choose a product probability reference $\mu = m_- \otimes m_+$ and density $r_S = \mathrm{d}R_S/\mathrm{d}\mu > 0$. Then

$$\hbar D_{\mathrm{KL}}(\gamma\|R_S) = \int c_S\,\mathrm{d}\gamma + \hbar D_{\mathrm{KL}}(\gamma\|\mu), \quad c_S = -\hbar \log r_S. \quad (13)$$

The cost is determined once the action, boundary maps, and reference measures are specified. Computing r_S may still require the interacting path integral; the static formulation does not remove that computational work. Normalized reference changes $m'_- \propto e^{-a/\hbar} m_-$ and $m'_+ \propto e^{-b/\hbar} m_+$ give $c'_S = c_S - a - b+$ a constant, with a compensating entropy change. Separately, adding $a(x)+b(y)$ to a cost at fixed μ changes a fixed-marginal objective by a constant. Supplementary Appendix S.3 gives the sign, normalization, and dual calculation.

The constrained optimizer is the original law only when its prescribed marginals already match those of P. Where positive scaling factors exist,

$$\frac{\mathrm{d}Q_\star}{\mathrm{d}P} = f(B_-)g(B_+), \qquad \mathbb{E}_P O = \mathbb{E}_{Q_\star}\frac{O}{f(B_-)g(B_+)}. \quad (14)$$

The reweighting is essential: differentiating an arbitrary fixed-marginal minimum does not reproduce Eq. (6). Neither this identity nor entropy minimization alone is a variance-reduction guarantee.

IV. SCATTERING CONNECTION AND FERMIONIC SCOPE

A. Continuation, scalar examples, and wavepackets

A regulated positive Euclidean law is not an S matrix. A continuum, infinite-volume limit satisfying the Osterwalder–Schrader hypotheses permits Lorentzian reconstruction [13, 14]. Isolated stable one-particle poles and the scattering assumptions then permit LSZ reduction of connected time-ordered correlators [15, 16]. The finite-dimensional variational identity proves neither these limits nor their numerical stability.

For the scattering conventions take $\hbar = c = 1$, metric $(+,-,-,-)$, and $S = I + iT$, with the reduced matrix element $\langle q|T|p\rangle = (2\pi)^4\delta^{(4)}(P_q - P_p)\mathcal{M}(q,p)$. For a scalar action with potential $m^2\phi^2/2 + g\phi^3/3! + \lambda\phi^4/4!$, take $\lambda > 0$ and a consistent vacuum expansion. For instance, $g^2 < 3\lambda m^2$ makes $\phi = 0$ a global classical minimum. The quadratic kernel is $p^2 - m^2$; the Feynman prescription, which is boundary data, selects $G_F = i/(p^2 - m^2 + i0)$. The action derivatives give vertices $-ig$ and $-i\lambda$. Wick contractions of the connected generator then produce

$$\mathcal{M}^{\mathrm{tree}}_{2\to 2} = -\lambda - g^2\Big[\frac{1}{s - m^2 + i0} + \frac{1}{t - m^2 + i0} + \frac{1}{u - m^2 + i0}\Big]. \quad (15)$$

The factorial cancellations and factors of i are derived in Supplementary Appendix S.5. At an isolated exchange pole the residue factorizes into the two cubic vertices and the propagator numerator. For equal external masses the $s = m^2$ pole lies below the physical $s \geq 4m^2$ region; it is an analytic pole, not one reached in the physical packet scan. Coupling order and loop order differ: $L = I - V + 1 = 0$ for the two-vertex exchange graph.

For smooth packet amplitudes, the transition part is

$$\mathcal{A}_\Psi = \int \mathrm{d}\Pi_p \mathrm{d}\Pi_q\, \Psi^*_{\mathrm{out}}(q)(2\pi)^4\delta^{(4)}(P_q - P_p) \times i\mathcal{M}(q,p)\Psi_{\mathrm{in}}(p). \quad (16)$$

Thus a constant contact contribution gives $\mathcal{A}_\Psi = -i\lambda\mathcal{I}_\Psi$, not automatically $-i\lambda$. The distribution is paired with packets before taking an absolute square. Positivity and marginals of a Euclidean bridge do not replace $S^\dagger S = I$; that property belongs to the reconstructed scattering theory.

B. Fermionic entropy projection after reconstruction

Grassmann integration is algebraic. With a fixed Berezin order and invertible fermion matrix D, completion of the Grassmann square gives

$$\int \mathcal{D}\bar{\psi}\mathcal{D}\psi\, e^{(-\bar{\psi}D\psi+\bar{\eta}\psi+\bar{\psi}\eta)/\hbar} = \det(D/\hbar)e^{\bar{\eta}D^{-1}\eta/\hbar}. \tag{17}$$

This identity does not make a negative or complex determinant a probability. Supplementary Appendix S.6 derives the ordered contractions and distinguishes them from operator positivity.

As a specified Euclidean starting class, consider the free-overlap chiral-Yukawa regulator of Ref. [5], including its auxiliary fermion and nearest-neighbor scalar action. That reference proves link reflection positivity for this interacting nongauge model. Its reflection maps $(t, \mathbf{x})$ to $(1-t, \mathbf{x})$ and exchanges ψ with the transposed Dirac-conjugate reflected field, reversing Grassmann order. After nonzero normalization, reflection positivity gives the OS form $(F, G) = \langle\Theta(F)G\rangle$ and quotient $\mathcal{H}_{a,\mathrm{OS}} = \overline{\mathcal{A}_+/\mathcal{N}}$. The local interaction does not require a positive pointwise fermion determinant.

Hamiltonian reconstruction additionally needs compatible time-translation and limiting assumptions [17, 18]. At nonzero time spacing translations are initially discrete. If they supply a nonnegative self-adjoint contraction T_a, define $H_a = -(\hbar/a_\tau)\log T_a$ on its positive spectral support, treating zero modes separately. We do not infer a continuous-time scattering theory from one finite periodic-lattice reflection inequality. Nor is the OS quotient automatically an asymptotic Fock space. These qualifications are detailed in Supplementary Appendix S.6.

Suppose a parity-even, lower-bounded reconstructed H_a is supplied and its lower bound is shifted to zero. Retain a nonzero finite-rank spectral subspace $P_{E_c} = \mathbf{1}_{[0,E_c]}(H_a)$. On it define

$$\Omega = Z^{-1}e^{-\beta H_{a,E_c}/\hbar}, \quad Z = \mathrm{Tr}\, e^{-\beta H_{a,E_c}/\hbar}, \quad \beta > 0. \tag{18}$$

Here β is Euclidean time, not an independently fitted transport cost. Every retained eigenvalue is strictly positive. Given specified trace-preserving completely positive maps $\mathcal{B}_\pm$, impose $\mathcal{B}_\pm(\Gamma) = \rho_\pm$, $\mathrm{Tr}\,\Gamma = 1$, $\Gamma \geq 0$, and parity invariance. The operator information-projection problem is

$$\min_{\Gamma\in\mathcal{C}} \hbar S(\Gamma\|\Omega), \qquad S(\Gamma\|\Omega) = \mathrm{Tr}\,\Gamma(\log\Gamma - \log\Omega). \tag{19}$$

For nonempty $\mathcal{C}$ the objective is continuous on a compact convex set and strictly convex, proving existence and uniqueness. This is a finite-matrix application of the quantum entropy framework [6, 7]. The state in Eq. (18) is a reference *density operator*, not by itself a two-time process operator. Equivalence of the constrained density projection to a scattering amplitude would need a further dynamical construction and is not asserted here. Commuting operators reduce to classical relative entropy, but that fact alone does not identify their prior with a particular path-space law.

For the separate perturbative Yukawa test, $\mathcal{L}_F = \bar{\psi}(i\partial\!\!\!/ - m_\psi)\psi + (\partial\phi)^2/2 - m_\phi^2\phi^2/2 - y\bar{\psi}\psi\phi$, action differentiation gives $-iy$ and $S_F(q) = i(q\!\!\!/ + m_\psi)/(q^2 - m_\psi^2 + i0)$. The scalar denominator carries the causal boundary prescription. Ordered contractions and fermionic LSZ give

$$i\mathcal{M}_{\psi\phi\to\psi\phi} = (-iy)^2\bar{u}(p')[S_F(p+k) + S_F(p-k')]u(p). \tag{20}$$

CAR gives a minus sign on exchanging two identical incoming legs or two identical outgoing legs. An incoming–outgoing interchange is instead a crossing operation. Equation (20) is a field-theory consistency test, not a numerical solution of Eq. (19).

V. NUMERICAL TESTS AND WHAT THEY ESTABLISH

All calculations use dimensionless parameters and double precision. The code, positive quadrature rules, seeds, raw arrays, and independent reference routines are supplied. The original one-mode bridge, contact extraction, scalar response, Yukawa, and packet plots are retained in the reproduction archive. Five selected figures distinguish the full-interaction comparison, the boundary solve, and the response/packet checks. The interacting three-site calculation and its low-order perturbative comparison are additions to the submitted version.

A. Nonperturbative finite-regulator scalar validation

Take three real field coordinates $\phi_1, \phi_2, \phi_3 \in [-L, L]$ with open-chain action

$$S_E = \frac{1}{2}\sum_{j=1}^{3}\phi_j^2 + \frac{\kappa}{2}\sum_{j=1}^{2}(\phi_{j+1} - \phi_j)^2 + \frac{\lambda}{24}\sum_{j=1}^{3}\phi_j^4, \tag{21}$$

where $\hbar = 1$, $\kappa = 0.7$, $L = 6$, and $\lambda \in \{0.5, 2, 4\}$. This is a three-site Euclidean scalar model with a field-amplitude cutoff, not a three-spatial-dimensional scattering simulation. Let $R_{0,h}$ be the normalized positive product-quadrature mass for the quadratic part and

TABLE I. Interacting three-site entropy solve. $\Delta F = -\log(Z_\lambda/Z_0)$; $C_{13} = \langle\phi_1\phi_3\rangle$; $\kappa_4 = \langle\phi_2^4\rangle - 3\langle\phi_2^2\rangle^2$. Values use $n = 80$, $L = 6$, and 100 mirror steps.

λ	ΔF	$\langle\phi_2^2\rangle$	C_{13}	κ_4
0.5	0.06737944	0.50253694	0.07232029	−0.02799851
2.0	0.21094894	0.42805664	0.04529325	−0.04691680
4.0	0.34379167	0.37529866	0.03063063	−0.04847402

$V_\nu = \lambda\sum_j \phi_{j,\nu}^4/24$. Tensor Gauss–Legendre grids use $n = 16, 24, 32, 48, 64, 80$ nodes per field coordinate; h denotes this quadrature regulator, not Planck's constant.

We optimize all n^3 probability masses with the full interaction retained:

$$F_h(q) = \sum_\nu q_\nu V_\nu + \sum_\nu q_\nu \log(q_\nu/R_{0,h,\nu}). \tag{22}$$

Starting at $q^{(0)} = R_{0,h}$, entropic mirror descent with step $\tau = 1/4$ gives

$$q_\nu^{(k+1)} \propto q_\nu^{(k)} e^{-\tau[V_\nu + \log(q_\nu^{(k)}/R_{0,h,\nu})+1]}. \tag{23}$$

Updates and normalization are evaluated in log space. No interacting target probability or amplitude is passed to the update. The finite-grid exponential solution is constructed afterward to check the Gibbs identity and excess objective. This elementary convex solve is not presented as a new optimization algorithm; it verifies the complete interacting objective rather than an imposed channel response.

Independent validation uses nested adaptive integration of Eq. (21), not the tensor grid. Conditional factorization reduces the partition function to $\int e^{-u(y)} b_0(y)^2 \mathrm{d}y$, where $u(x) = x^2/2 + \lambda x^4/24$ and $b_j(y) = \int_{-L}^{L} x^j e^{-u(x)-\kappa(x-y)^2/2}\mathrm{d}x$. The endpoint correlator uses $b_1(y)^2$ instead. Supplementary Appendix S.8 derives the reference integrals, gradient residual, and refinement criteria. The optimizer's variance and connected fourth cumulant are not fixed by normalization.

Table I gives the 80^3-mass result after 100 iterations. Against the independent integration, all three free-energy differences are below 10^{-12}, relative central-variance differences below 10^{-11}, and absolute connected-fourth-cumulant differences below 10^{-11}. These are observed cross-method differences, not rigorous interval bounds. At $\lambda = 2$, changing L from 6 to 7 changes the reference free energy by 7.3×10^{-13} or less. The separate optimization convergence at $n = 48$ is shown in Supplementary Fig. S1; Fig. 1 instead compares physical moments at different couplings.

a. Comparison with low-order perturbation theory. Write $S_E = S_0 + \lambda U$, with $U = \sum_j \phi_j^4/24$. All branches use the same cutoff $[-6, 6]^3$. The normalized identity

$$\langle O\rangle_\lambda = \frac{\langle O e^{-\lambda U}\rangle_0}{\langle e^{-\lambda U}\rangle_0} \tag{24}$$

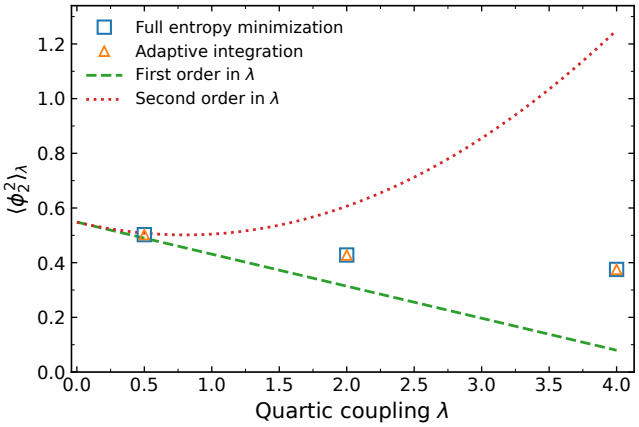


FIG. 1. Nonperturbative finite-regulator scalar validation. Hollow squares show the full entropy minimization and hollow triangles independent adaptive integration at three couplings; the symbols coincide to the displayed precision. Curves are the first- and second-order Taylor truncations about $\lambda = 0$ for the same cutoff theory. No nonperturbative curve is interpolated between the computed points. The full quartic interaction suppresses the central variance in the tested model; low-order truncations depart substantially at the larger couplings.

defines first- and second-order Taylor truncations by re-expanding the ratio, not by dividing two truncated series without re-expansion. Supplementary Appendix S.8 gives the coefficients and their $n = 80$ to 96 refinement check. The full entropy solve retains the exponential interaction rather than either truncation.

Figure 1 and Table II compare the central variance. At $\lambda = 2$ the first- and second-order relative discrepancies are 26.6% and 41.7%; at $\lambda = 4$ they are 78.8% and 233%. The full calculation agrees with adaptive integration to the displayed digits. For the connected fourth cumulant at $\lambda = 2$, the full value is −0.04691680, while its consistently re-expanded second-order value is +0.34909577 (Supplementary Fig. S2). Also $U \geq 0$ implies $\Delta F \geq 0$ for $\lambda \geq 0$, whereas the second-order free-energy truncation is −0.16815104 at $\lambda = 4$, compared with the full value 0.34379167. These comparisons establish accuracy beyond the tested low-order truncations for this finite model. They do not compare with higher-order or resummed perturbation theory, establish an OT-specific speedup, or constitute a nonperturbative scattering prediction.

B. Boundary bridge for the same interacting action

At $\lambda = 2$, $n = 48$, prescribe endpoint quadrature masses proportional to $w_i \exp[-(x_i + 0.50)^2/(2\,0.70^2)]$ and $w_k \exp[-(x_k - 0.45)^2/(2\,0.65^2)]$. The middle-site marginal is not prescribed. Log-space alternating scaling is applied to the three-index action weight $u_{ijk} = R_{0,h,ijk} e^{-V_{ijk}}$. This solves the constrained version of

TABLE II. Central variance for the same nonperturbative comparison. Full entropy minimization and adaptive integration agree to all eight displayed decimal places. The two perturbative columns are independently truncated through the stated order.

λ	Full / reference	First order	Second order
0.5	0.50253694	0.48979113	0.50807862
2.0	0.42805664	0.31400324	0.60660309
4.0	0.37529866	0.07961939	1.25001880

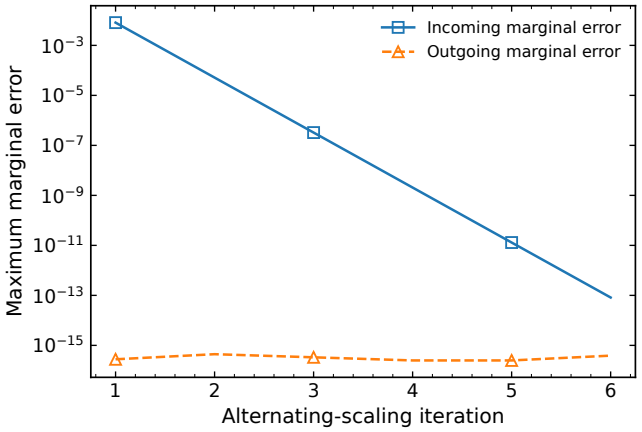


FIG. 2. Boundary entropy projection of the same interacting three-site law. Hollow squares and triangles give maximum incoming and outgoing marginal errors. The update enforces the outgoing marginal last, so that error is already near round-off while the incoming error decreases. The original and constrained ensembles have different endpoint correlators; their relation is verified by reweighting, not by equating the measures.

Eq. (22), retaining the interacting conditional distribution of the middle field.

Figure 2 shows convergence of both endpoint constraints. They are satisfied after six scaling sweeps to maximum errors 8.23×10^{-14} and 3.89×10^{-16}. The path and endpoint relative entropies both equal approximately 0.51272751; their difference is below 10^{-14}, checking Eq. (10) with unchanged conditional laws. The endpoint correlator changes from 0.04529325 in P_h to -0.18882489 in the prescribed bridge. This change is expected, not a failure: the boundary constraints change the ensemble. Dividing out the endpoint scalings and renormalizing recovers the original finite-grid masses to maximum absolute discrepancy below 10^{-15}. No sampling-variance advantage is inferred from this deterministic reweighting check.

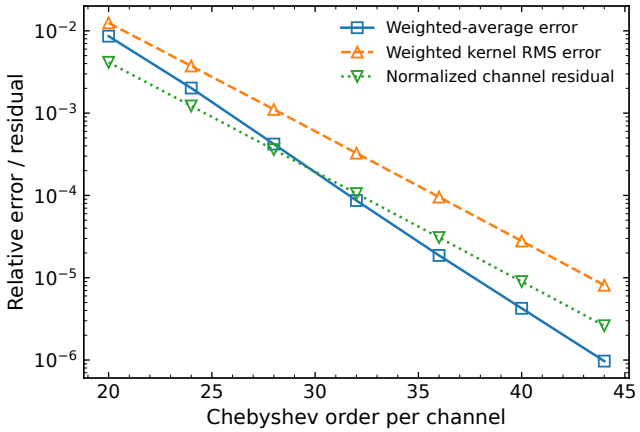


FIG. 3. Spectral channel reconstruction, evaluated on eight three-dimensional packet configurations. Hollow squares show the largest weighted-average error, upward triangles the largest weighted kernel RMS error, and downward triangles the largest normalized channel residual. Three functions of one invariant are fitted using Eq. (25); no bridge is optimized in this test.

C. Scalar and fermionic response checks

The scalar channel check solves $D(x)K(x)+ig^2=0$, with $D=x-m^2+i\eta$, using an order-N Chebyshev expansion and complex least squares. For three channels it minimizes

$$J_N=|K_4+i\lambda|^2+\sum_{c=s,t,u}\sum_k w_{c,k}\times|D(x_{c,k})K_c^{(N)}(x_{c,k})+ig^2|^2. \tag{25}$$

The inverse-propagator factor is supplied from the action. Closed solution values are withheld until the coefficients are fixed. Supplementary Appendix S.7 derives the normal equations and the exact error relation $K_N-K_\star=R_N/D$. This numerical problem is not a minimization over path probabilities or a transport coupling, and it is not equivalent to Eq. (22) by definition.

For $m=1$, $g=0.55$, $\lambda=0.30$, $\eta=0.03$, three one-variable channel functions are tested on eight three-dimensional packet geometries with central momenta 0.72–1.80 and angles 25°–150°. There are $4N$ collocation nodes per channel. The sampled extrema set the channel intervals, but the packet amplitudes are not training labels. Figure 3 retains the original refinement test with its narrower interpretation. At $N=44$, the worst weighted-average discrepancy is 9.71×10^{-7} or less, the worst weighted kernel RMS discrepancy is 8.12×10^{-6} or less, and the normalized channel residual is below 2.62×10^{-6}. The values certify channel reconstruction, not an OT-specific gain or a full six-dimensional variational field solve.

The Yukawa check evaluates Eq. (20) by direct inversion of the Dirac matrix and independently through its closed numerator/denominator expression. For $m_\psi=1$,

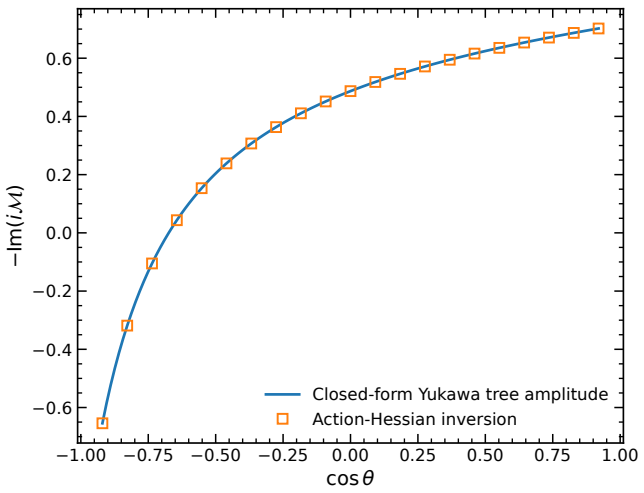


FIG. 4. Yukawa tree response. The curve uses the closed propagator expression; hollow squares use numerical Dirac-matrix inversion. Parameters and spin labels are specified in the text. Agreement checks Eq. (20); the calculation does not solve the operator information projection of Eq. (19).

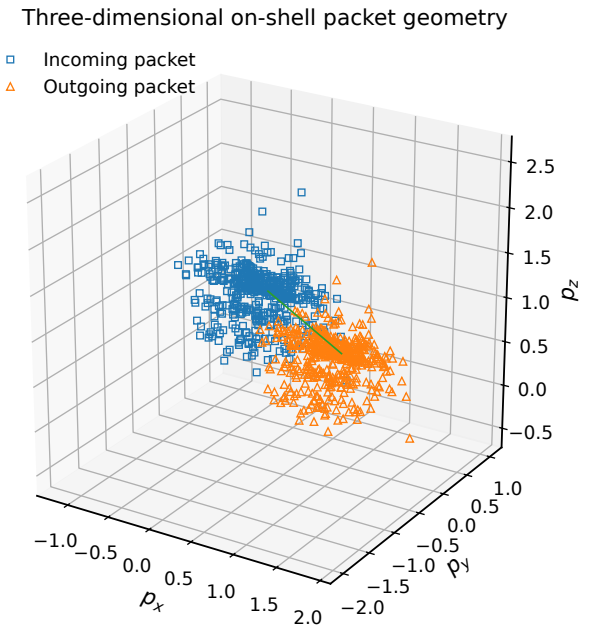


FIG. 5. Retained three-dimensional momentum-packet geometry. Hollow squares and triangles sample the incoming and outgoing anisotropic Gaussian factors. The centers have magnitude 1.05 and relative angle 58°. This visualization describes the finite-resolution six-dimensional packet integration, not the three-site entropy model. Total-momentum reduction and the finite energy window are specified in the text and Supplementary Appendix S.7.

$m_\phi = 0.4$, $\sqrt{s} = 3.2$, $y = 0.6$, and equal canonical spin labels, the two agree at the roundoff scale; the maximum relative difference in the archived run is below 10^{-12}. Figure 4 compares them. It tests Dirac algebra and spinor conventions, not interacting fermionic entropy minimization.

D. Three-dimensional packets and the six-dimensional integral

Figure 5 restores the three-dimensional geometry of the retained packet benchmark to the main article. The calculation integrates over two reduced center-of-mass momenta $\mathbf{p}, \mathbf{q} \in \mathbb{R}^3$ with $\mathrm{d}\nu_m = \mathrm{d}^3\mathbf{p}/[(2\pi)^3 4E_\mathbf{p}^2]$. It uses anisotropic Gaussian packet factors and a Gaussian replacement for the energy delta of width $\sigma_E = 0.18$ with $\eta = 0.03$. This is a *finite-resolution integration test.* Finite packet width alone does not relax exact energy conservation of the infinite-time S matrix. Away from equal energies its invariant extension obeys $s + t + u = 4m^2 + 4E_\mathbf{p}(E_\mathbf{p} - E_\mathbf{q})$ rather than $4m^2$. Supplementary Appendix S.7 derives this distinction. The convergence plot and configuration-by-configuration data remain in the reproduction archive.

The nested 2^{20} and 2^{21} Sobol evaluations differ by about 8.50×10^{-6} in their kernel-average ratio and 4.45×10^{-4} in the unnormalized integral. They are convergence differences relative to a finer run, not independent total-error bounds or a demonstrated $\sigma_E \to 0$ limit. The eight-geometry response comparison above uses its archived positive energy-window weighting, not an invariant cross-section measure. These qualifications preserve the numerical results while identifying which physical and numerical conclusions they support.

VI. DISCUSSION AND CONCLUSIONS

The action-based QOTAM representation separates three levels. The unconstrained Gibbs minimum gives the regulated generating functional and its connected correlators. Boundary constraints give a Schrödinger information projection that generally changes the ensemble. Scattering amplitudes require Lorentzian reconstruction and asymptotic reduction, not the identification of a positive coupling with a complex kernel. The derivations appendix supplies the Gaussian, entropy, source, scalar, and fermionic algebra behind these statements.

The interacting three-site calculation verifies an important distinction in practice. It minimizes the full non-Gaussian probability objective without a coupling expansion, reproduces independently integrated moments, and constructs a nontrivial endpoint bridge. Reweighting then recovers the original law. The coupling comparison additionally shows agreement with independent integration where the tested first- and second-order moment approximations are inaccurate, including a second-order cumulant with the wrong sign at $\lambda = 2$. This is a finite-regulator nonperturbative validation of the entropy rep-

resentation, not a statement about all perturbative methods. It does not establish efficient large-lattice computation, nonperturbative four-dimensional scattering, or superiority over direct integration or conventional sampling. The scalar/Yukawa and multidimensional packet checks serve complementary, explicitly more limited purposes.

For fermions, the reference Gibbs operator follows from a specified reconstructed Hamiltonian and regulator data. The finite density-matrix information projection is well posed when its affine constraints are feasible. Reflection positivity and Hamiltonian reconstruction must be justified for the chosen Euclidean regulator; a sign-changing determinant is not converted into a classical probability. A universal two-time scattering interpretation of the constrained operator problem and a numerical solution of its interacting fermionic version remain separate tasks.

Perturbative expansions are available from the same generating functional, while its exact finite-regulator identity is nonperturbative. Neither statement proves regulator removal or an efficient solver. Higher connected functions allow finite higher-multiplicity reductions when scattering hypotheses hold. Tensor or learned representations may approximate the admissible measures or density matrices, but compression and computational advantage require independent tests. The present results therefore establish a documented finite-regulator formulation and reproducible checks, with the transport, correlator, and scattering steps kept mathematically distinct.

DATA AND CODE AVAILABILITY

The accompanying reproduction package contains the code, fixed seeds, positive quadrature rules, numerical arrays, figure sources, and run records used for the reported tests. The interacting calculation, low-order comparison coefficients, and retained earlier response calculations are identified separately. No external experimental data are used. The detailed derivations are provided in the accompanying appendix.

Appendix A: Core variational derivations

This appendix gives the algebra behind the principal identities in the article. The longer supplementary appendix supplies normalization integrals, field-theory contractions, operator variations, and numerical details.

a. Gibbs identity. For $U_J = V_N - J\cdot\Phi$, the normalized tilt in Eq. (2) gives, on the stated finite-entropy, finite-moment domain,

$$\log\frac{\mathrm{d}Q}{\mathrm{d}P_{N,J}} = \log\frac{\mathrm{d}Q}{\mathrm{d}R_{0,N}} + \frac{U_J}{\hbar} + \log\mathcal{Z}_N[J]. \tag{A1}$$

Integrating against Q yields Eq. (4). The interacting law is admissible by the assumed moment condition, so relative-entropy nonnegativity proves attainment and uniqueness. For two source-independent splits, the full partition integral $Z_{0,A}\mathcal{Z}_A[J] = Z_{0,B}\mathcal{Z}_B[J]$ is the same; therefore the difference between the variational objectives is independent of both Q and J.

b. Disintegration and boundary reweighting. With $T = (B_-, B_+)$, write $Q(\mathrm{d}\Phi) = Q^{xy}(\mathrm{d}\Phi)\gamma(\mathrm{d}x\mathrm{d}y)$ and $P(\mathrm{d}\Phi) = P^{xy}(\mathrm{d}\Phi)R_S(\mathrm{d}x\mathrm{d}y)$. The density factorization

$$\frac{\mathrm{d}Q}{\mathrm{d}P}(\Phi) = \frac{\mathrm{d}\gamma}{\mathrm{d}R_S}(T\Phi)\frac{\mathrm{d}Q^{xy}}{\mathrm{d}P^{xy}}(\Phi) \tag{A2}$$

implies Eq. (10) after taking logarithms and expectations. Minimizing the conditional term retains P^{xy}. If $\mathrm{d}\gamma_\star/\mathrm{d}R_S = fg$, then $\mathrm{d}Q_\star/\mathrm{d}P = f(B_-)g(B_+)$, which proves the observable reweighting in Eq. (14). These are the standard Schrödinger reduction identities [3].

c. Transport Lagrangian and dual. For $q = \mathrm{d}\gamma/\mathrm{d}\mu$, $\mu = m_-\otimes m_+$ a product probability measure, introduce the two marginal multipliers α, β. The Lagrangian is

$$\begin{aligned} L = &\int [c_S q + \hbar q\log q - (\alpha+\beta)q]\mathrm{d}\mu \\ &+ \int \alpha\mathrm{d}\rho_- + \int \beta\mathrm{d}\rho_+. \end{aligned} \tag{A3}$$

Its pointwise stationary density and exact dual function are

$$\begin{aligned} q_\star &= \exp[(\alpha+\beta-c_S)/\hbar - 1], \\ D(\alpha,\beta) &= \int \alpha\mathrm{d}\rho_- + \int \beta\mathrm{d}\rho_+ \\ &\quad - \hbar\int e^{(\alpha+\beta-c_S)/\hbar - 1}\mathrm{d}\mu. \end{aligned} \tag{A4}$$

The -1 is required for the entropy convention $q\log q$. For normalized reference changes $m'_- = Z_a^{-1}e^{-a/\hbar}m_-$ and $m'_+ = Z_b^{-1}e^{-b/\hbar}m_+$ at fixed R_S, one has $r'_S = Z_aZ_be^{(a+b)/\hbar}r_S$. Therefore $c'_S = c_S - a - b - \hbar\log(Z_aZ_b)$; the entropy contribution changes oppositely.

d. Source derivatives and the operator identity. Differentiating the normalized tilt, subject to the stated domination assumption, gives

$$\partial_{J_i}P_{N,J}(\mathrm{d}\Phi) = \frac{\Phi_i - \langle\Phi_i\rangle_J}{\hbar}P_{N,J}(\mathrm{d}\Phi). \tag{A5}$$

It implies $\partial_{J_i}\langle O\rangle_J = \mathrm{Cov}_J(O,\Phi_i)/\hbar$ and hence Eq. (6). For the finite-matrix state, $\log\Omega = -\beta H/\hbar - (\log Z)I$ gives

$$\hbar S(\Gamma\|\Omega) = \beta\,\mathrm{Tr}\,\Gamma H + \hbar\,\mathrm{Tr}\,\Gamma\log\Gamma + \hbar\log Z. \tag{A6}$$

Continuity on the finite density-matrix set and strict convexity of the entropy term give the constrained existence/uniqueness statement in Sec. IV. This does not construct a two-time process or supply a new scattering theorem.

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